Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/31/2006
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Chung Law's email is cklaw@princeton.edu		
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	01/01/2003	**End Date:**	12/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: this grant on NCE until 12/31/2007, per K. Sacksteder/GRC (11/07) NOTE: grant to be reinstated with a new grant number. 8/2006, per D. Urban		
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):	Tse, Stephen (Rutgers University) Sacksteder, Kurt (NASA Glenn Research Center)		
Grant/Contract No.:	NCC3-1051		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the GRC and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
Task Progress:	Previously, the structure and extinction of diffusion flames generated by a porous spherical burner or a fuel droplet in response to rotational motion were investigated through perturbation analysis, with emphasis on the effects of non-unity Lewis numbers (Le) for both fuel and oxidizer. Constant density was assumed for simplicity of analysis, hence allowing decoupling of the flow from the thermal effects. Solutions were found for all relevant quantities, and explicit expressions were given for the burning rate and the distorted flame shape. While the constant density assumption simplifies the analysis considerably by decoupling the flow from the thermal effects and provides elegant analytical results, such simplification in the analysis is known to give an incorrect flame location compared to experimental results, which might possibly lead to over- or under-prediction the effects of rotation. To address this, we carried out the analysis without making such simplification and the solutions were obtained numerically. While it was shown that main feature of the problem was adequately described by the constant-density model, the variable-density formulation revealed two new insights: (1) perturbations due to rotation decrease substantially as compared with the constant-density formulation, suggesting that the perturbing effects of rotation are substantially absorbed and thereby mitigated by the density variation, (2) magnitude of the perturbations strongly depends on the ratio of the burner/droplet surface temperature to the ambient temperature. Finally, microgravity experiments were conducted at NASA Glenn not only to verify the theoretical predictions but also to extend or supplement the theories to study the flame response at high angular velocities where the perturbation theory no longer applies. The experiments were conducted using both the 2.2 second drop tower and the 5 second zero-G facility at GRC. The flames were generated by injecting either the fuel or the oxidizer mixtures through a bronze porous burner to a controlled ambient which also comprises either an oxidizer or fuel mixture. In the low angular velocity limit where the perturbation analysis predictions are valid, no visible changes were observed, as the perturbations predicted by the theory were also small. In the other extreme, the effect of rotation was clearly visible and the response agreed surprisingly well for most conditions with the perturbation theory. The scaling for the deformation of the flame was also analyzed separately at the poles and the equator. In particular, the polar displacement of the flame as a function of the angular velocity scaled similarly as that established by the spinning disk.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Yoo SW, Qian J, Bechtold JK, Law CK. "Response of spherical diffusion flames under rotation with general Lewis numbers." Combustion Theory and Modeling, 2005 May;9(2):199-217. , May-2005